MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, GENERAL PURPOSE

TYPES 1N3189, 1N3190, 1N3191, TX AND NON-TX TYPES

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon, general purpose semiconductor diodes for use as power rectifiers in equipment circuits and is in accordance with MIL-S-19500, except as otherwise specified herein. The prefix "TX" is used on devices which have been submitted to and have passed the special process-conditioning, testing, and screening as specified in 4.4 through 4.4.5.3.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

Type	V_R	I_O T_A = 150°C	I_O T_A = 100°C	I_{FSM} 1/120 sec
	Vdc	Adc	Adc	A(pk)
1N3189	200	0.5	1	30
1N3190	400	0.5	1	30
1N3191	600	0.5	1	30

OPERATING JUNCTION TEMPERATURE: -65° to +175°C

STORAGE AMBIENT TEMPERATURE: -65° to +175°C

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements for the diodes shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The diode shall be of the design, construction, and physical dimensions specified on figure 1.

3.3.1 Lead finish. Lead finish shall be gold plated or tinned. Lead finish may be specified in the contract or order (see 6.2) without affecting the qualified product status of the device or applicable JAN marking.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I and II (table III is applicable to "TX" types only).

3.4.1 Process-conditioning, testing, and screening for "TX" types. Process-conditioning, testing, and screening for the "TX" types shall be in accordance with MIL-S-19500 and as specified in 4.4.

3.5 Marking. The marking shall be as specified in MIL-S-19500 with the following exceptions:

(a) Manufacturer's identification, and country of origin may be omitted at the option of the manufacturer.
(b) "TX" devices (all): The type designation may be abbreviated by using the "JX" prefix in lieu of the "JANTX" prefix.

3.5.1 It is permissible to have the type designation on more than one line. Examples of acceptable marking are as follows:

JAN1N 3189	or	JX1N 3189

3.5.2 Polarity. The polarity shall be indicated by a diode graphic symbol with the arrow pointing toward the negative terminal for forward bias.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I and II. Group A inspection shall be performed on an inspection sublot of each type. Subgroups 1,2,3, and 4 of group B inspection may be performed on an inspection sublot of any type to qualify all types. Subgroup 5 of group B inspection shall be performed on the highest-voltage type within a barometric-pressure group to qualify that type and all lower-voltage types within the barometric-pressure group. Subgroups 6 and 7 of group B inspection shall be performed on an inspection sublot. Subgroups 8 and 9 of group B inspection shall be performed on an inspection sublot of the highest- and lowest-voltage types being qualified.

4.2.1 Instability. The device shall be swept through the reverse characteristic to the peak reverse voltage and the trace displayed on an oscilloscope. The device shall, at the same time, be subjected to shocks of 50 G's minimum at a minimum rate of 20 shocks per second for a minimum of 2 seconds or acceptable alternative. Any instability noted in the reverse trace shall be considered a failure.

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4.3 Quality conformance inspection. Quality conformance inspection shall consist of examinations and tests specified in groups A and B inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I. Group A inspection shall be conducted on an inspection sublot of each type.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II. Group B inspection may be done on a lot basis with the exception of subgroups 8 and 9. For subgroups 8 and 9, sublots of the highest-voltage types present in the lot shall be used to accept those types and all intermediate-voltage types.

4.4 Process-conditioning, testing, and screening for "TX" types. The procedure for process-conditioning, testing, and screening the "TX" types shall be in accordance with 4.4.1 through 4.4.5.3.

4.4.1 High-temperature storage. All devices shall be stored at least 48 hours at a minimum temperature (TA) of 175°C.

4.4.2 Thermal shock (temperature cycling). Thermal shock (temperature cycling) shall be performed in accordance with MIL-STD-750, method 1051, test condition C, except that 10 cycles shall be continuously performed and the time at the temperature extremes shall be 15 minutes, minimum.

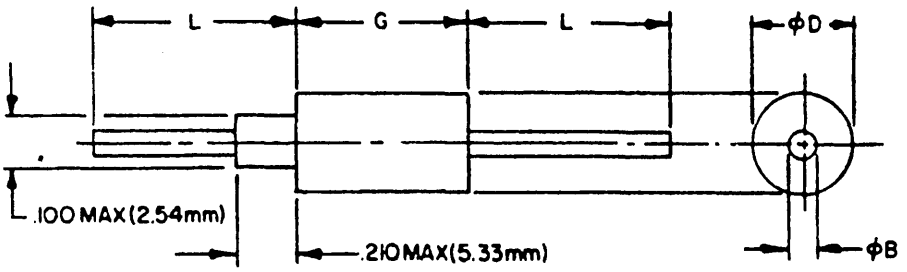
4.4.3 Acceleration. All devices shall be subjected to an acceleration test in accordance with MIL-STD-750, method 2006, with the following exceptions: The test shall be performed one time in the y_1 orientation only at a peak level of 20,000 G minimum. The 1-minute hold-time requirement shall not apply.

4.4.4 Hermetic seal tests. All devices shall be subjected to hermetic seal tests (fine leak followed by gross leak).

4.4.4.1 Hermetic seal (fine-leak) test. All cavity devices shall be fine-leak tested in accordance with MIL-STD-750, method 1071, test condition G or H.

4.4.4.2 Hermetic seal (gross-leak) test (bubble). All devices shall be tested for gross leaks in accordance with MIL-STD-750, method 1071, test condition D, except that the solution may be any suitable noncorrosive liquid at a minimum temperature of 100°C.

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LTR	DIMENSIONS			
	INCHES		MILIMETERS	
	MIN	MAX	MIN	MAX
ØB	.026	.041	.66	1.04
ØD	.195	.230	4.95	5.84
G		.350		8.89
L	1.000		25.40	

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
2. Metric equivalents are in parentheses.
3. Cathode lead shall be electrically connected to the case. If tubulation is used, it shall be on the anode end.

FIGURE 1. Semiconductor device, diode 1N3189, 1N3190, 1N3191, TX, and non-TX types.

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TABLE I. Group A inspection.

Examination or test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TX	TX		Min	Max	
Subgroup 1			5	5				
Visual and mechanical	2071				---	---	---	---
Subgroup 2			5	1				
Forward voltage	4011	I_F = 750 mAdc			V_F	---	1	Vdc
Reverse current	4016							
1N3189		V_R = 200 Vdc			I_R	---	5	µAdc
1N3190		V_R = 400 Vdc			I_R	---	5	µAdc
1N3191		V_R = 600 Vdc			I_R	---	5	µAdc
Reverse current	4016							
1N3189		V_{RM} = 240 V(pk)			I_{RM}	---	100	µA(pk)
1N3190		V_{RM} = 480 V(pk)			I_{RM}	---	100	µA(pk)
1N3191		V_{RM} = 720 V(pk)			I_{RM}	---	100	µA(pk)
Instability		See 4.2.1						
Subgroup 3			5	3				
High temperature operation		T_A = 150°C			---	---	---	---
Reverse current	4016							
1N3189		V_R = 200 Vdc			I_R	---	500	µAdc
1N3190		V_R = 400 Vdc			I_R	---	500	µAdc
1N3191		V_R = 600 Vdc			I_R	---	500	µAdc

TABLE II. Group B inspection.

Examination or test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TX	TX		Min	Max	
Subgroup 1			20	20				
Physical dimensions	2066				---	---	---	---
Subgroup 2			15	10				
Solderability	2026				---	---	---	---
Temperature cycling	1051	Test condition F1			---	---	---	---
Thermal shock	1056	Test condition A			---	---	---	---
Moisture resistance	1021				---	---	--	---

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TABLE II. Group B inspection - Continued.

Examination or test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TX	TX		Min	Max	
Subgroup 2 - continued								
End points:								
Forward voltage	4011	I_F = 750 mAdc			V_F	---	1.0	Vdc
Reverse current	4016							
1N3189		V_R = 200 Vdc			I_R	---	5.0	µAdc
1N3190		V_R = 400 Vdc			I_R	---	5.0	µAdc
1N3191		V_R = 600 Vdc			I_R	---	5.0	µAdc
Subgroup 3			15	10				
Shock	2016	Nonoperating; 1,500G, 5 shocks 0.5 ms 2 major axes			---	---	---	---
Constant acceleration	2006	10,000G (non-operating)			---	---	---	---
Vibration, variable frequency	2056	Nonoperating			---	---	---	---
End points: (Same as subgroup 2)								
Subgroup 4			15	10				
Terminal strength (lead fatigue)	2036	Test condition E			---	---	---	---
Subgroup 5			15	15				
Barometric pressure, reduced	1001							
1N3189		V_R = 200 Vdc			---	---	---	---
1N3190		V_R = 400 Vdc			---	---	---	---
1N3191		V_R = 600 Vdc			---	---	---	---
		Altitude = 70,000 feet						
Salt atmosphere	1041				---	---	---	---
End points: (Same as subgroup 2)								

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TABLE II. Group B inspection - Continued.

Examination or test	MIL-STD-750		LTPD		Symbol	Limits		Unit
	Method	Details	Non-TX	TX		Min	Max	
Subgroup 6			15	5				
Surge current	4066	I_{FSM} = 30A(pk) I_F = 250 mAdc 5 surges, 1 surge/min			---	---	---	---
End points: (Same as subgroup 2)								
Subgroup 7			15	15				
Peak reverse voltage		T_A = -65°C						
1N3189		V_{RM} = 240 V(pk)			I_{RM}	---	100	µA(pk)
1N3190		V_{RM} = 480 V(pk)			I_{RM}	---	100	µA(pk)
1N3191		V_{RM} = 720 V(pk)			I_{RM}	---	100	µA(pk)
Subgroup 8			λ=10	λ=5				
Steady state operation life	1026	I_O = 1.0 Adc T_A = 100° ±3°C						
1N3189		V_R(RMS) = 140 Vac			---	---	---	---
1N3190		V_R(RMS) = 280 Vac			---	---	---	---
1N3191		V_R(RMS) = 420 Vac			---	---	---	---
End points:								
Forward voltage	4011	I_F = 750 mAdc			V_F	---	1.1	Vdc
Reverse current	4016							
1N3189		V_R = 200 Vdc			I_R	---	10.0	µAdc
1N3190		V_R = 400 Vdc			I_R	---	10.0	µAdc
1N3191		V_R = 600 Vdc			I_R	---	10.0	µAdc
Subgroup 9			λ=10	λ=5				
High temperature life (non-operating)	1031	T_A = 175° ±3°C			---	---	---	---
End points: (Same as subgroup 8)								

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4.4.5 Preburn-in tests. The parameters I_R and V_F of table III shall be measured and the data shall be recorded for all devices in the lot. All devices shall be handled and identified such that the delta end points can be determined after the burn-in test. All devices which fail to meet these requirements initially shall be removed from the inspection lot and the quantity removed shall be noted on the permanent lot history.

TABLE III. Burn-in test measurements.

Test	MIL-STD-750		Symbol	Limits		Unit
	Method	Details (T_A = 25° ±3°C)		Min	Max	
Reverse current	4016	1N3189; V_R = 200 Vdc	I_R	---	5.0	µAdc
		1N3190; V_R = 400 Vdc	I_R	---	5.0	µAdc
		1N3191; V_R = 600 Vdc	I_R	---	5.0	µAdc
Forward voltage	4011	I_F = 750 mAdc	V_F	---	1.0	Vdc

4.4.5.1 Burn-in test. All devices shall be operated for 96 hours minimum under the following conditions:

T_A = 100°C minimum

V_R = 140 Vac(rms) - 1N3189
280 Vac(rms) - 1N3190
420 Vac(rms) - 1N3191

I_o = 1.0 Adc

4.4.5.2 Post-burn-in tests. The parameters I_R and V_F of table III shall be retested after burn-in and the data shall be recorded for all devices in the lot. The parameters measured shall not change during the burn-in test from initial value by more than the following amount:

ΔI_R = 100 percent or 2 µAdc, whichever is greater.

ΔV_F = ±0.025 Vdc.

4.4.5.3 Burn-in test failures. All devices that exceed the delta (Δ)limits of 4.4.5.2 or the limits of table III after burn-in shall be removed from the inspection lot and the quantity removed shall be noted on the permanent lot history. Where the quantity removed after burn-in exceeds 10 percent of the total inspection lot on burn-in test, the entire lot shall be unacceptable. Acceptable devices shall then be tested in accordance with the requirements of group A inspection (table I) and group B inspection (table II).

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Ordering data. Procurement documents should specify the following:

Lead finish if preference exists (see 3.3.1).

User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

Agent:
DSA - ES

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